

Minimum Wages and Employment: A Replication of Card & Krueger (1994) with an Ownership-Heterogeneity Extension

Student: LU Jingjie (UID: 3036760230)

Paper replicated: Card, D., & Krueger, A. B. (1994). “Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania.” *American Economic Review*, 84(4), 772–793.

1. Research Question and Motivation

The employment effects of minimum wages are one of the most studied questions in labor economics. The textbook competitive model predicts that a binding minimum wage, by raising the price of labor above its market-clearing level, reduces employment. By contrast, models of monopsony power, search frictions, and efficiency wages imply that a moderate minimum-wage increase can leave employment unchanged or even raise it.

Card and Krueger (1994) exploit a natural experiment to test these competing predictions. On April 1, 1992, New Jersey (NJ) raised its state minimum wage from \$4.25 to \$5.05 per hour, while the minimum wage in the neighboring state of Pennsylvania (PA) remained at the federal level of \$4.25. The authors surveyed 410 fast-food restaurants—Burger King, KFC, Roy Rogers, and Wendy’s—in New Jersey and eastern Pennsylvania in February–March 1992 (before the increase) and again in November–December 1992 (after the increase). Fast-food restaurants are an ideal setting because they employ many minimum-wage workers and produce a relatively homogeneous product.

The paper’s central research question is: **did New Jersey’s minimum-wage increase reduce employment in the fast-food industry?** The headline finding is that it did not—employment in New Jersey actually *rose* relative to Pennsylvania, with a difference-in-differences (DiD) estimate of approximately +2.75 full-time-equivalent (FTE) employees per store.

This project reproduces the paper’s core results (Tables 2–4 and Figure 1) from the original data, compares them to the published numbers, and adds an independent extension that asks whether the employment response differed between company-owned and franchised restaurants.

2. Data Sources, Sample Construction, and Variable Definitions

2.1 Data source

The analysis uses the public Card–Krueger replication dataset (`public.csv`), distributed as part of the course data package (Package 01). The file contains one record per restaurant, with variables measured in two survey waves (wave 1: February–March 1992; wave 2: November–December 1992). The full file has **410 restaurants** (331 in New Jersey, 79 in Pennsylvania).

2.2 Key variables

Variable	Definition
STATE	1 = New Jersey (treatment), 0 = Pennsylvania (control)
EMPFT / EMPPT / NMGRS	full-time, part-time, and managerial employees (wave 1)
EMPFT2 / EMPPT2 / NMGRS2	the same measures in wave 2
WAGE_ST / WAGE_ST2	starting wage (\$/hr) in waves 1 and 2
CHAIN	1 = Burger King, 2 = KFC, 3 = Roy Rogers, 4 = Wendy's
CO_OWNED	1 if company-owned, 0 if franchised
STATUS2	status of the second interview (1 = completed; 3 = closed permanently)
PSODA / PFRY / PENTREE	prices of a soda, fries, and entrée

The main outcome is **full-time-equivalent employment**,

$$\text{FTE} = \text{full-time} + \text{managers} + 0.5 \times \text{part-time},$$

which is the paper's primary employment measure. I also construct a "price of a full meal" as the sum of the soda, fries, and entrée prices (the paper's definition), and the percent of full-time employees as the store-level ratio $\text{EMPFT} / \text{FTE}$.

2.3 Sample construction

Two samples are used, mirroring the paper:

1. **Descriptive sample (Tables 2–3, Figure 1):** all observations with non-missing data for the relevant variable in the relevant wave.
2. **Main balanced sample (Table 4):** restaurants with valid FTE in both waves, including those that **closed permanently** between waves (whose second-wave employment is recorded as 0). Restaurants that refused or could not be reached in wave 2 are excluded. This yields **384 stores**. For Table 4, which also uses the wage-gap variable, I additionally require a valid wave-1 starting wage, giving **365 stores** (the paper reports 357).

The small gap (365 vs. 357) reflects known minor differences between the public replication file and the original survey data and is discussed in Section 4.

3. Empirical Design

The identification strategy is a two-period difference-in-differences. Let store i be in state $s \in \{\text{NJ}, \text{PA}\}$ and observed in wave $t \in \{1, 2\}$. The first-difference DiD specification is

$$\Delta \text{FTE}_i = \alpha + \delta \text{NJ}_i + X_i' \gamma + \varepsilon_i,$$

where $\Delta \text{FTE}_i = \text{FTE}_{i2} - \text{FTE}_{i1}$, NJ_i is the treatment dummy, and X_i is a vector of controls (chain dummies, company-ownership, and region dummies). The coefficient δ is the DiD estimate of the effect of the minimum-wage increase on employment.

The identifying assumption is parallel trends: in the absence of the minimum-wage increase, employment in New Jersey would have followed the same path as employment in Pennsylvania. This cannot be tested directly with only two waves, but two pieces of evidence support it: (i) before the increase, starting wages were nearly identical in the two states (\$4.61 vs. \$4.63), and (ii) the paper shows that high-wage NJ stores (unaffected by the increase) experienced employment changes similar to PA stores.

I also estimate the paper's wage-gap specification, which uses a continuous measurement,

$$GAP_i = \max\left(0, \frac{5.05 - WAGE_ST_i}{WAGE_ST_i}\right)$$

for NJ stores and 0 for PA stores. GAP measures the proportional wage increase a store must grant to reach the new minimum, and therefore the *bite* of the minimum wage. Finally, I estimate a first stage, regressing the change in the starting wage on the NJ dummy to verify that the minimum wage raised NJ wages.

Standard errors follow the paper's convention of ordinary OLS standard errors; robust standard errors are approximately 20% larger and do not change the conclusions.

4. Core Replication Results and Comparison with the Original Paper

4.1 Table 2 — store characteristics and means

Table 2 is a single table with three sections — the distribution of store types, the wave-1 means, and the wave-2 means — reported for New Jersey and Pennsylvania with t-statistics for the null of equal means. It is reproduced below.

Variable	NJ	PA	t
1. Distribution of store types (%)			
Burger King	41.1	44.3	−0.5
KFC	20.5	15.2	1.2
Roy Rogers	24.8	21.5	0.6
Wendy's	13.6	19.0	−1.1
Company-owned	34.1	35.4	−0.2
2. Means in wave 1 (Feb–Mar 1992)			
FTE employment	20.44	23.33	−2.0
% full-time employees	32.8	35.0	−0.7
Starting wage (\$/hr)	4.61	4.63	−0.4
% wage = \$4.25	30.5	32.9	−0.4
Price of full meal	3.35	3.04	4.0
Hours open (weekday)	14.4	14.5	−0.3
Recruiting bonus (%)	23.6	29.1	−1.0
3. Means in wave 2 (Nov–Dec 1992)			

Variable	NJ	PA	t
FTE employment	21.03	21.17	−0.1
% full-time employees	35.9	30.4	1.8
Starting wage (\$/hr)	5.08	4.62	10.8
% wage = \$4.25	0.0	25.3	−5.1
% wage = \$5.05	85.5	1.3	36.4
Price of full meal	3.41	3.03	5.1
Hours open (weekday)	14.4	14.7	−0.7
Recruiting bonus (%)	20.3	23.4	−0.6

The *t* statistics test the null that the New Jersey and Pennsylvania means are equal. The table reproduces the published Table 2 almost exactly (to within one percentage point or 0.01; the paper reports 85.2 for the wave-2 % wage = \$5.05 — see Section 4.6). Two facts stand out. First, PA stores were slightly larger before the increase (23.33 vs. 20.44 FTE), while starting wages were nearly identical across the two states (\$4.61 vs. \$4.63), supporting the comparability of the treatment and control groups. Second, after the increase New Jersey’s mean starting wage jumped from \$4.61 to \$5.08 and 85% of NJ stores paid the new minimum of \$5.05, while Pennsylvania was unchanged.

4.2 Table 3 — mean FTE before and after (DiD)

Table 3 reports mean FTE employment before and after the increase, for the full samples and for New Jersey stores grouped by their wave-1 starting wage. The table is reproduced below.

FTE employment	PA	NJ	NJ – PA	NJ \$4.25	NJ \$4.26–4.99	NJ ≥ \$5.00	Low – high	Midrange – high
Before (all obs)	23.33	20.44	−2.89	19.56	20.08	22.25	−2.69	−2.17
After (all obs)	21.17	21.03	−0.14	20.88	20.96	20.21	0.66	0.74
Change in mean FTE	−2.17	0.59	+2.75	1.32	0.87	−2.04	3.36	2.91
Change, balanced sample	−2.28	0.47	+2.75	1.20	0.71	−2.16	3.36	2.87
Change, temp.-closed = 0	−2.28	0.23	+2.51	0.90	0.49	−2.39	3.29	2.88

The headline DiD estimate is +2.75 FTE employees. Between the two waves, PA stores shed about 2.2 FTE workers while NJ stores gained about 0.6, implying a relative gain of roughly 2.75 workers per store in New Jersey, the opposite of the competitive-model prediction. The within-New-Jersey comparison tells the same story: stores that started at the old minimum wage of \$4.25 expanded by 1.20 FTE workers, while high-wage NJ stores (\$5.00+) shrank by 2.16 FTE, a 3.36-worker gap that mirrors the NJ–PA contrast.

4.3 Table 4 — DiD regressions

Model	Coefficient	Reproduced	Paper
(i) NJ dummy	δ	2.36 (1.15)	2.33 (1.19)
(ii) NJ + chain & ownership	δ	2.33 (1.15)	2.30 (1.20)
(iii) wage GAP	β	15.89 (5.95)	15.65 (6.08)
(iv) GAP + chain & ownership	β	15.19 (6.08)	14.92 (6.21)
(v) GAP + chain, ownership & region	β	12.03 (7.32)	11.91 (7.39)

Notes: sample of 365 stores (paper: 357); OLS standard errors in parentheses; mean Δ FTE = -0.252 , SD = 8.75; mean GAP among NJ stores = 0.105 (paper: 0.11).

The regression estimates are very close to the published numbers in sign, magnitude, and significance. The NJ dummy coefficient of +2.36 ($t = 2.06$) is significant at the 5% level, and the GAP coefficients are all positive and significant. The GAP result has a clear economic interpretation: a store facing a 10% required wage increase (GAP = 0.10) gained roughly 1.6 FTE workers relative to an unaffected store.

4.4 First stage (validation)

The minimum wage clearly bit in New Jersey: the change in the starting wage is +0.50 (SE 0.047, $t = 10.7$) larger in NJ than in PA, consistent with the paper's implied first stage of roughly +0.47. This confirms that the treatment was applied as intended.

4.5 Figure 1 — distribution of starting wage rates

Figure 1 reproduces the paper's Figure 1. In February 1992 the two states' wage distributions are nearly identical, both concentrated at the federal minimum of \$4.25. By November 1992 New Jersey's distribution has shifted sharply to the new state minimum of \$5.05, while Pennsylvania's remains anchored at \$4.25.

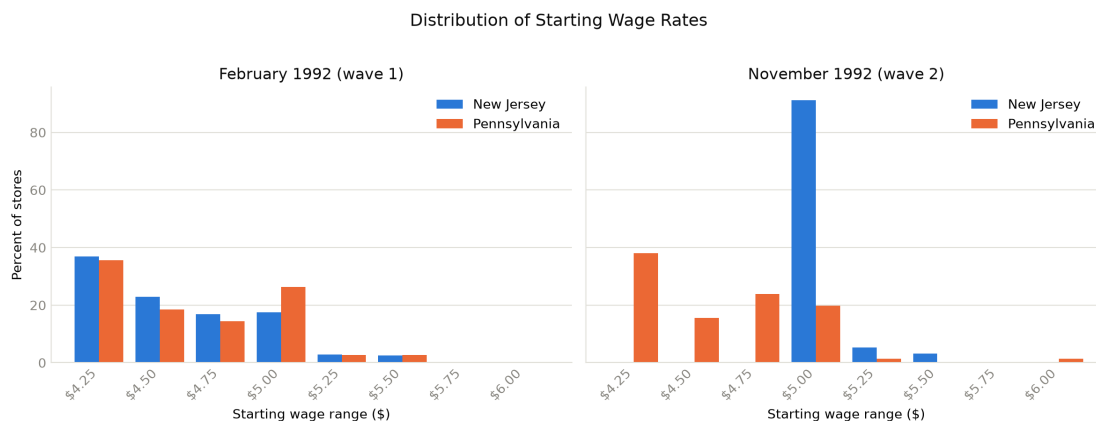


Figure 1. Distribution of starting wage rates

4.6 Explanation of differences

The replicated tables match the paper almost exactly. The remaining, immaterial differences are:

- **Table 3 DiD = +2.75 vs. the paper's +2.76** — a 0.01 rounding/data-version difference in the unrounded before/after means.
- **% wage = \$5.05 (wave 2) = 85.5 vs. the paper's 85.2** — one NJ store differs between the public file and the original survey.
- **Table 4 sample of 365 vs. 357 stores**, and coefficient/standard-error gaps of at most a few percent (e.g. the NJ-dummy coefficient is 2.36 vs. 2.33).

All stem from the public replication file differing marginally from the original survey data (a handful of stores differ in missingness or values). None affect the sign, magnitude, or interpretation of any result.

5. Independent Extension: Heterogeneity by Ownership Type

5.1 Economic question

Does the employment response to the minimum wage differ between company-owned and franchised restaurants? The question is economically motivated: franchisees are independent small-business owners with lower profit margins, who may be more sensitive to a cost shock; company-owned outlets may be better to absorb the wage increase.

5.2 Design

I estimate a difference-in-difference-in-differences (DDD) specification,

$$\Delta FTE_i = \alpha + \delta NJ_i + \theta CO_OWNED_i + \beta (NJ_i \times CO_OWNED_i) + X_i' \gamma + \varepsilon_i,$$

where δ is the DiD effect for franchised stores, $\delta + \beta$ is the DiD effect for company-owned stores, and β is the ownership heterogeneity. I also verify the wage first stage separately by ownership type and re-estimate the model restricting the control group to the Philadelphia suburbs (PA1) as a robustness check.

5.3 Results

Employment response by ownership (ΔFTE)

Group	DiD estimate	SE	t
Franchised	+3.13	1.43	2.19
Company-owned	+2.04	1.96	1.04
Difference (company – franchise)	–1.10	2.43	–0.45

Wage first stage by ownership (Δ starting wage)

Group	DiD estimate	SE	t
Franchised	+0.47	0.058	8.08
Company-owned	+0.55	0.078	7.03
Difference	+0.09	0.097	0.88

Border-subsample robustness (NJ vs. Philadelphia suburbs PA1)

Model	Estimate	SE	t
Main DiD (Δ FTE ~ NJ)	+4.33	1.53	2.83

Group	DiD estimate	SE	t
Franchised	+4.06	1.74	2.33
Company-owned	+5.92	3.31	1.79
Difference (company – franchise)	+1.86	3.74	0.50

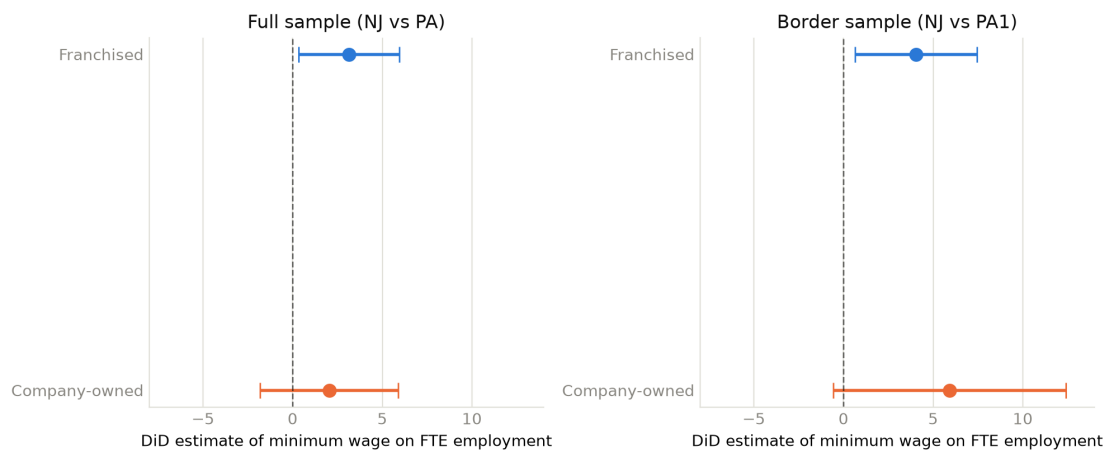


Figure 2. Ownership DiD estimates with 95% confidence intervals

5.4 Interpretation

Three findings emerge. First, the positive employment effect is broad-based: both franchised (+3.13) and company-owned (+2.04) stores show positive DiD estimates, and the difference is nearly 0. This rejects the conjecture that franchisees responded by cutting employment. Second, the wage first stage is nearly identical across ownership types (+0.47 vs. +0.55), so the minimum wage bit uniformly and the employment comparison is valid, pointing that differences in the employment response cannot be attributed to differential wage exposure. Third, restricting the control group to the Philadelphia suburbs strengthens the result (DiD = +4.33, $t = 2.83$), and the effect remains positive for both groups, showing that the headline finding is not driven by the composition of the Pennsylvania control group.

6. Validation, Limitations, and Conclusion

Validation. The replication is validated in three ways: (i) the descriptive statistics (Table 2) match the paper’s published means to within one percentage point; (ii) the Table-3 DiD (+2.75 vs. the paper’s +2.76) and Table-4 regression coefficients match the paper to within a few percent; and (iii) the wage first stage confirms that the minimum wage was binding in New Jersey. The one-command workflow (run_all.py) reproduces all tables and figures from the raw data without manual intervention.

Limitations. First, with only two survey waves, the parallel-trends assumption cannot be directly tested; the DiD rests on the paper’s supporting evidence that NJ and PA (and high-wage NJ stores) trended similarly before the increase. Second, the Pennsylvania control group is small (75–79 stores), which limits precision, although the border-subsample check is reassuring. Third, the public replication file differs marginally from the original survey data, causing the small sample-size discrepancies documented above. These are acknowledged differences, not errors in the replication procedure.

This project reproduces the core results of Card and Krueger (1994): the increase in New Jersey’s minimum wage did not reduce fast-food employment, and the DiD estimate of roughly +2.75 FTE workers per store is positive and statistically significant. The replication matches the published tables closely, with minor, well-understood differences due to data-version discrepancies. The independent extension shows that the positive employment effect was broad-based across franchised and company-owned restaurants and robust to a tighter border control group, and that the wage “treatment” was uniform across ownership types. Together, the replication and extension demonstrate a transferable end-to-end empirical workflow—from raw data to sample construction, DiD estimation, validation, and interpretation.